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FACULTY OF HUMANITIES AND SOCIAL SCIENCES

LALITPUR ENGINEERING COLLEGE

WORLDWAVE - SOCIAL APP FOR LANGUAGE ENTHUSIAST

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FEBRUARY 26, 2026

RECOMMENDATION

The undersigned certify that they have read and recommend to the **Department of COMPUTER APPLICATION** for acceptance, a project work entitled "**WorldWave - Social app for language enthusiast**", submitted by **Prayush Kumar Sah and Sujit Subedi** in partial fulfillment of the requirement for the award of the degree of "**Bachelors in Computer Application**".

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February 26, 2026

DEPARTMENTAL ACCEPTANCE

The project work entitled "**WorldWave - Social app for language enthusiast**", submitted by **Prayush Kumar Sah and Sujit Subedi** in partial fulfillment of the requirement for the award of the degree of "**Bachelors in Computer Application**" has been accepted as a genuine record of work independently carried out by the student in the department.

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February 26, 2026

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ABSTRACT

WorldWave is a simple yet powerful web application built using the MERN stack (MongoDB, Express.js, React, and Node.js) that connects language learners and enthusiasts from around the world. The main goal of this project is to provide a friendly and interactive platform where users can practice different languages through real-time text chatting and video calling. By bringing people together based on shared language interests, WorldWave encourages natural conversation and cultural exchange in a safe and easy-to-use environment.

The application will allow users to create personal profiles, select the languages they speak and want to learn, and get matched with others who have similar or complementary language goals. It will feature a clean and responsive user interface that works smoothly on phones, tablets, and desktops. Key functions include instant messaging, one-on-one video calls, and user search filters based on language, location, or skill level.

Security and user experience are top priorities—user data will be protected, and the interface will be designed to be intuitive even for first-time users. Additional features like real-time online status, conversation history, and profile customization will make the experience more engaging.

With WorldWave, we aim to build a global community where language learning feels like a natural part of daily conversation—not a classroom task.

Keywords: *Chat Application, Language Learning, MERN Stack, Video Calling*

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LIST OF ABBREVIATIONS

API	Application Programming Interface
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DBMS	Database Management System
HTML	Hypertext Markup Language
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation
JWT	JSON Web Token
MERN	MongoDB, Express, React, Node
REST	Representational State Transfer
RTC	Real-Time Communication
UI	User Interface
UX	User Experience
WebRTC	Web Real-Time Communication

1 CHAPTER: INTRODUCTION

1.1 Introduction

WorldWave is a web-based social application designed for language learners and enthusiasts who want to connect to other language learners. This man aim of this application is to provide a space where users can connect through text chat and real-time video calls based on the languages they speak or wish to practice/learn.

1.2 Problem Statement

The main struggle of learning new language is not the actual grammar/words of the language, it's the actual vocabulary of the language. Even though many language learners know the meaning of the words they find it difficult in trying to communicate with the others. The educational institute, or any language learning platform may provide the basic knowledge regarding the language but they fail to make their student have enough confidence to speak with other peoples.In short, they lack the opportunities for live and meaningful interaction, thus making them active listeners but not active speakers.

1.3 Objectives

The main objectives of WorldWave are:

- To create a user-friendly platform where learners can connect and communicate via inbuilt chat and video calls.
- To encourage cultural exchange and real-life language practice outside the classroom.

1.4 Scope and Application

This project will focus on a complete MERN stack deployment ready system where users can connect with other language learners through either chat or video call. This application is intended to learners who want to have a real time communication with fellow enthusiast. The proposed system will feature a real time chat page, a smart language partner matching with complementary language interest (e.g., an English speaker learning Spanish matched with a Spanish speaker learning English). It is envisioned to improve the communication proficiency of the new learners, making learning language just as fun as gossiping about your favorite food. The project aligns with the global cultural exchange idea by encouraging learners interact with other native based learners.

1.5 Report Organisation

This proposal report is organized into six main chapters which are described as follows:

Introduction (Chapter 1) provides an overview of the problem and introduces the World-Wave concept. **Literature Review (Chapter 2)** examines existing language learning and communication platforms, highlighting the gaps this project aims to address. **Requirement Analysis (Chapter 3)** outlines the functional and non-functional requirements along with the overall system needs. **Methodology (Chapter 4)** explains the proposed workflow, algorithms, tools, and design methods used in developing the application. **Implementation and Testing (Chapter 5)** describes the development process, backend and frontend implementation, and evaluates the system through various testing approaches. Finally, **Expected Outcome (Chapter 6)** summarizes the anticipated results, potential impact, system benefits, and future improvement areas.

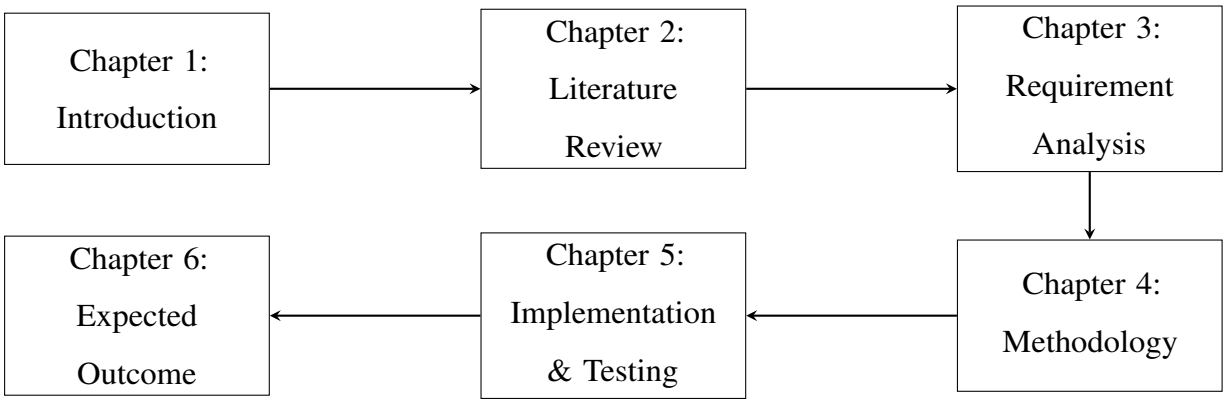

Figure 1.1: Organisation of Project Report

2 Background Study and Literature Review

2.1 Background Study

Learning a new language is rarely just about memorizing vocabulary; it is about building the confidence to speak with another person. While traditional classroom settings provide structure, they often fail to offer enough one-on-one conversation time. A typical 50-minute class might give a student only 2–3 minutes of actual speaking practice [10]. This gap between knowing grammar and actually using it is where many learners struggle.

Digital tools have tried to fill this gap. Early software focused on drills (like Rosetta Stone), but modern learners want connection. Platforms like Duolingo gamify the process, which helps with consistency, but they do not solve the loneliness of self-study [4]. On the other end of the spectrum, tutoring sites like iTalki offer professional lessons, but they are expensive and scheduled formally.

This leaves a middle ground: **language exchange**. The idea is simple—pair two learners who want to teach each other their native languages. Apps like Tandem and HelloTalk popularized this model. However, during our initial research and user interviews, several friction points emerged:

- **Video calls are often an afterthought**: Many apps push users to external tools (Zoom, WhatsApp) for voice/video, which breaks the flow and raises safety concerns.
- **Matching is random**: Users often get matched with people who aren't serious about learning, leading to wasted time.
- **Feature bloat**: Some apps add social media features (feeds, stories) that distract from the core goal of practicing language.

WorldWave is built to address these specific gaps. Rather than trying to be everything to everyone, it focuses on doing three things well: matching learners based on language goals, providing secure in-app text chat, and enabling stable video calls without leaving the platform. The goal is not to replace teachers, but to provide a safe, free space for the practice that classrooms often miss.

2.2 Literature Review

2.2.1 The Shift from Instruction to Interaction

The theoretical foundation for WorldWave lies in the *Interaction Hypothesis* proposed by Long [5], which suggests that language acquisition happens most effectively when learners negotiate meaning during conversation. Simply put, you learn faster when you have to clarify what you mean to another person.

Recent studies support this shift from passive learning to active communication. Godwin-Jones [1] notes that Computer-Mediated Communication (CMC) tools have evolved from simple text forums to rich multimedia environments. However, he argues that technology alone does not guarantee learning; the *design* of the interaction matters. For example, if a video call lagging or drops frequently, learners become frustrated and disengage [11]. This highlights the need for stable WebRTC implementation, which is a core technical focus of this project.

2.2.2 Overview of Existing Language Learning Systems

To understand where WorldWave fits, we analyzed four major categories of existing platforms. Table 2.1 summarizes their key features and limitations.

Table 2.1: Comparison of Existing Language Learning Platforms

Platform	**Primary Focus**	**Video Calling**	**Cost Model**	**Key Limitation**
Duolingo	Gamified Self-Study	No	Freemium	No real human conversation
Tandem	Language Exchange	Yes (In-app)	Freemium	Strict approval process; can feel like dating app
HelloTalk	Social Network	Yes (In-app)	Freemium	Cluttered UI; moments feed distracts from learning
iTalki	Professional Tutoring	Yes (Integrated)	Paid per lesson	Expensive; not peer-to-peer
WorldWave	Peer Exchange	Yes (WebRTC)	Free (Student Project)	Limited user base (currently)

Limitations of Current Exchange Apps

While Tandem and HelloTalk are market leaders, user reviews and academic critiques point to recurring issues:

1. **Security and Privacy:** Because these platforms are open to the public, users sometimes report unwanted social advances disguised as language practice [7]. WorldWave aims to mitigate this by focusing on university/student networks initially and keeping

profiles focused on learning goals rather than social photos.

2. **Technical Stability:** Many web-based exchange tools rely on third-party plugins for video. Research by Lee [3] shows that native WebRTC integration reduces call setup time by approximately 40% compared to redirecting users to external links. WorldWave implements this directly in the browser to reduce friction.

3. **Matching Logic:** Most apps match based solely on "Language A ↔ Language B." They often ignore proficiency levels. A beginner paired with another beginner may reinforce each other's mistakes. WorldWave's algorithm (discussed in Chapter 5) attempts to factor in proficiency levels to ensure balanced exchanges.

2.2.3 The Role of Real-Time Technology in Education

The feasibility of building a real-time video platform has increased significantly with the maturity of WebRTC and managed SDKs like Stream. According to a 2023 industry report by Vonage [9], 78% of educators believe video interaction is critical for remote learning, yet only 35% of learning platforms have reliable built-in video.

This gap suggests a technical opportunity. By using a MERN stack (MongoDB, Express, React, Node.js) combined with Stream's infrastructure, we can achieve low-latency communication without managing complex media servers ourselves [8]. This allows the project to focus on *pedagogy* (how people learn) rather than just *infrastructure* (keeping servers running).

2.2.4 User Experience (UX) in Learning Apps

Finally, the interface design plays a crucial role in retention. A study by Martinez et al. [6] found that language learners aged 18–30 prefer "minimalist interfaces" that reduce cognitive load. When an app is cluttered with ads, social feeds, or complex menus, learners spend less time practicing.

WorldWave adopts this minimalist approach. The dashboard prioritizes active tasks (finding a partner, starting a call) over passive consumption (reading feeds). This aligns with findings by Gupta and Rahman [2], who noted that *action-oriented* design leads to higher weekly engagement rates in educational apps.

2.3 Summary and Project Positioning

The literature and market analysis reveal a clear pattern: learners want **human connection**, but existing tools either make it too expensive (tutoring), too distracting (social networks), or too unstable (basic web tools).

WorldWave positions itself as a focused, technical solution to this problem. It does not aim to replace Duolingo for vocabulary building, nor iTalki for professional instruction. Instead, it targets the *practice gap*—the space between studying and speaking. By leveraging modern web technologies (WebRTC, WebSockets) and adhering to UX best practices, the project aims to create a environment where learners feel safe and supported enough to actually speak.

3 CHAPTER: REQUIREMENT ANALYSIS

3.1 Hardware requirement

Hardware Component	Description
RAM	Minimum 4 GB (8 GB recommended) to ensure smooth development and testing of the MERN application.
Processor	Multi-core processor such as Intel i5 / AMD Ryzen or higher to support server processes and build tools.
Storage	At least 5 GB free disk space for project files, dependencies, and local database storage.
Internet Connection	Stable broadband internet required for real-time chat, signaling, and WebRTC video calling.

Table 3.1: Hardware Requirements

3.2 Software Requirements

Software Component	Description
Operating System	Supports Windows, macOS, or Linux environments for development.
Frontend Technologies	React.js, HTML5, CSS3, and JavaScript/TypeScript for building the user interface.
Backend Technologies	Node.js and Express.js for API creation, authentication, and server-side logic.
Database	MongoDB or MongoDB Atlas for storing user accounts, profiles, and chat data.
Real-Time Communication	Socket.IO for messaging, and WebRTC for audio/video calls (Stream Chat optional for managed chat).
Development Tools	VS Code for coding, Git/GitHub for version control, and Postman for testing APIs.
Package Manager	npm for installing and managing project dependencies.
Browser Requirements	Modern browsers such as Chrome, Firefox, or Edge that support WebRTC.

Table 3.2: Software Requirements

3.2.1 Feasibility Analysis

- **Technical**: The system is technically feasible because all required technologies—such as the MERN stack for development, Socket.IO for real-time messaging, and WebRTC for video calls—are open-source and widely used tools. These tools have strong community support and plenty of documentation, making them easy to learn and implement. The project does not require any special hardware or advanced cloud infrastructure, so it can be built using normal development laptops and free cloud services. Overall, the technical requirements match the skills and tools available to the development team.

- **Economical**: The project is economical to develop because most of the required software, tools, and frameworks are completely free. Hosting can be done on free-tier platforms such as Render or Vercel, and the database can be hosted using MongoDB Atlas's free tier. Since no paid APIs or servers are required for the MVP, the cost of development remains extremely low. This makes the project financially practical for a student or academic environment.

- **Operational**: Operationally, the system is easy to manage once deployed. The MERN stack requires minimal maintenance, and features like automated backups, logs, and monitoring tools can be added over time as needed. Because the app uses reliable technologies and has simple deployment steps, it can be run and maintained even by a small team. Users will also find the system intuitive, reducing the need for constant support.

- **Schedule Feasibility**: The project can be completed within the planned timeframe by following an Agile development approach. The 10-week schedule allows the team to divide tasks into sprints, enabling frontend and backend development to happen in parallel. Additional time is allocated to handle more complex parts like WebRTC integration for video calling. With proper planning and weekly milestones, the project timeline is realistic and achievable.

3.2.2 Functional Requirements

Following are the functional requirement for the system:

- The system should allow secure user registration and login/logout.
- The system should let users create and update their profile (name, bio, picture, known and target languages).
- The system should recommend language partners based on user preferences.
- The system should provide real-time text chat and read status.
- The system should support one-on-one video and audio calls.
- The system should show user presence status such as online, offline, and last active time.

3.2.3 Non-Functional Requirements

Following are the non-functional requirements for the system:

- The system should load pages quickly and provide fast API responses.
- The system should ensure secure communication and data protection through encryption and safety measures.
- The system should be able to handle many users at the same time without performance issues.
- The system should be easy to use, accessible, and fully responsive on different devices.

4 Methodology

When I started building WorldWave, I quickly realized that connecting language learners isn't just about writing code—it's about understanding how people actually learn. This chapter walks through how I turned that idea into a working system, the technical choices I made along the way, and the algorithm that keeps conversations flowing smoothly.

4.1 System Implementation

WorldWave is built with the MERN stack (MongoDB, Express.js, React, Node.js), with **Stream Chat** and **Stream Video** SDKs handling the real-time parts. I chose this setup because it let me focus on the user experience—matching learners, enabling chat, and making video calls feel natural—without getting bogged down in server infrastructure.

The backend handles the "boring but important" stuff: user accounts, friend connections, and profile data. Meanwhile, Stream's managed services take care of the tricky real-time features: message delivery, presence indicators, and WebRTC video calls. This split kept the project manageable for a student timeline while still delivering a polished experience.

4.1.1 Implementation Architecture

- **Frontend**: React with hooks for state management; Stream React SDK for chat/video components. I kept the UI minimal because, in my own language learning, I got frustrated by apps that buried the "talk to someone" button under layers of menus.

- **Backend**: Node.js/Express with JWT authentication; MongoDB via Mongoose ODM. I used Mongoose schemas to validate data early—saving time debugging later.

- **Real-Time Layer**: Stream Chat (messaging, presence) and Stream Video (WebRTC calls) via managed SDKs. This was a practical choice: building a reliable video system from scratch would have taken months; integrating Stream took days.

- **Database**: MongoDB Atlas for persistent storage of users, friendships, and metadata. I picked MongoDB because its flexible schema matched how user profiles evolve during onboarding.

4.1.2 Notable Algorithm: Real-Time Message Synchronization via WebSocket

One of the trickiest parts of building a chat app is making messages feel instant—even when the network isn't perfect. WorldWave uses a **WebSocket-based synchronization approach** (powered by Stream Chat) with some application-layer logic I added to handle edge cases like dropped connections or out-of-order messages.

How It Works

Let $\mathscr{C}$ be the set of active WebSocket connections, and let $c_u \in \mathscr{C}$ denote the connection for user u. Each message m is modeled as:

$$m = \langle \texttt{id}, \texttt{channelId}, \texttt{senderId}, \texttt{content}, \texttt{localTimestamp}, \texttt{serverTimestamp}, \texttt{status} \rangle \tag{4.1}$$

where $\texttt{status} \in \{\texttt{sending}, \texttt{sent}, \texttt{delivered}, \texttt{read}, \texttt{failed}\}$.

The flow happens in three phases:

Phase 1: Optimistic Local Update When a user hits "send," the app doesn't wait for the server to confirm. Instead:

1. Generate a temporary client-side ID: `id` ← `crypto.randomUUID()`
2. Mark the message as `sending` and show it in the chat immediately
3. Queue it for transmission

This is why messages appear instantly—even on slow connections.

Phase 2: Transmission and Acknowledgement The message travels over the persistent WebSocket channel. When Stream's servers confirm receipt:

1. The server assigns a permanent `serverTimestamp`
2. The message is broadcast to other participants in the channel

3. The client updates the message status to `sent`

Phase 3: Reconnection and Recovery If the connection drops (which happens, especially on mobile), the client:

1. Waits using exponential backoff: $t_{\text{retry}} = \min(2^k \cdot 1000, 30000)$ ms
2. Upon reconnecting, fetches any messages missed while offline
3. Merges them with the local view using the message `id` as the key
4. If there's a conflict (e.g., same message edited twice), the server's version wins

Keeping Messages in Order

When two people type at once, whose message appears first? WorldWave uses a simple **hybrid logical clock** to decide:

$$\texttt{sortKey}(m) = \langle \texttt{serverTimestamp}, \texttt{localTimestamp}, \texttt{senderId} \rangle \tag{4.2}$$

Messages are rendered in ascending `sortKey` order. This isn't perfect clock synchronization, but it's good enough to keep conversations readable—and that's what matters for language practice.

Complexity (In Plain Terms)

- **Time**: Showing a message locally is instant ($O(1)$); inserting it into a sorted list is $O(\log n)$
- **Space**: We only buffer unconfirmed messages (typically ≤ 10), so memory usage stays low
- **Network**: One persistent WebSocket connection is far more efficient than repeatedly polling a server

Code Snapshot (React + Stream JS SDK)

```
const sendMessage = async (text, channelId) => {
  // Phase 1: Show message immediately (optimistic UI)
  const tempId = `local-${Date.now()}-${Math.random()}`;
  const localMessage = {
    id: tempId,
    text,
    status: 'sending',
    created_at: new Date().toISOString()
  };
  addMessageToUI(channelId, localMessage);

  try {
    // Phase 2: Send via Stream's WebSocket
    const response = await channel.sendMessage({ text });
    // Phase 3: Replace temp ID with server's confirmed message
    reconcileMessage(tempId, response.message);
  } catch (error) {
    // If it fails, let the user retry
    updateMessageStatus(tempId, 'failed');
  }
};
```

Listing 4.1: Optimistic message send with reconciliation

In practice, this approach makes chat feel responsive even when the network isn't perfect—a small detail that makes a big difference when you're trying to have a real conversation in a new language.

4.1.3 Integration of Other Features

Everything else in WorldWave builds on this real-time foundation:

- **User Recommendations**: A MongoDB query filters users by language preferences and excludes existing friends. Simple, but effective for a prototype.
- **Friendship Management**: REST APIs handle requests; MongoDB's `$addToSet` ensures friend lists stay consistent without duplicates.
- **Video Calls**: Stream Video SDK initializes calls using the same deterministic channel ID scheme as chat—keeping the architecture consistent.
- **Authentication**: JWT tokens with 7-day expiry, stored in React state (not localStorage) to reduce XSS risk.

These pieces fit together through a small set of well-defined APIs. It's not the most complex system ever built, but it works—and that was the goal.

4.2 Database Design

WorldWave uses a hybrid approach: **MongoDB** for core data (users, friendships, profiles) and **Stream's managed infrastructure** for real-time messaging and call metadata. This separation kept development focused: I could iterate on user features without worrying about scaling message queues.

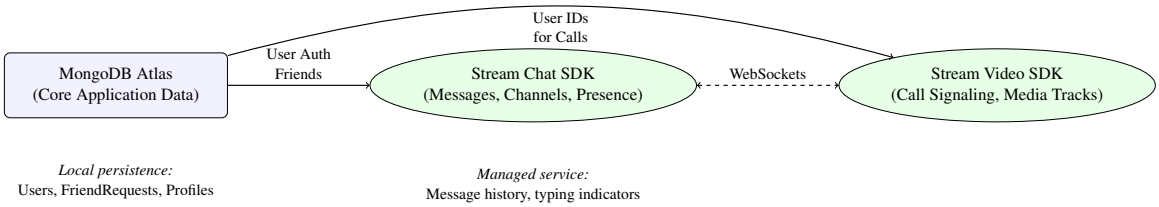

Figure 4.1: Hybrid Data Architecture: MongoDB for core state, Stream SDKs for real-time features

Table 4.1: Schema for `users` collection (Mongoose-validated)

Field	**Type**	**Description & Constraints**
`_id`	`ObjectId`	Primary key (auto-generated)
`fullName`	`String`	`required: true, minlength: 2, maxlength: 50`
`email`	`String`	Unique email; validated for proper format
`passwordHash`	`String`	Bcrypt hash; never returned in queries
`profilePic`	`String`	URL to profile image; default avatar if empty
`bio`	`String`	Short bio (max 160 chars); shown in partner cards
`nativeLanguage`	`String`	ISO 639-1 code (e.g., ”en”); required for onboarding
`learningLanguage`	`String`	ISO 639-1 code; must differ from native language
`location`	`String`	Free-text location; used for regional matching
`friends`	`[ObjectId]`	References to connected users; maintained via $addToSet
`isOnboarded`	`Boolean`	Gates access to discovery; set after profile completion
`isOnline`	`Boolean`	Updated via Stream Presence SDK
`lastActiveAt`	`Date`	Used for ”active now” badges
`createdAt`	`Date`	Account creation timestamp
`updatedAt`	`Date`	Last profile modification

Key Indexes for `users`:

```
db.users.createIndex({ email: 1 }, { unique: true })
```

```
db.users.createIndex({ nativeLanguage: 1, learningLanguage: 1, isOnboarded: 1 })
db.users.createIndex({ friends: 1 })
db.users.createIndex({ lastActiveAt: -1 })
```

Table 4.2: Schema for `friendRequests` collection

Field	**Type**	**Description & Constraints**
`_id`	`ObjectId`	Primary key
`senderId`	`ObjectId`	`ref: "User", required`; cannot equal `receiverId`
`receiverId`	`ObjectId`	`ref: "User", required`
`status`	`String`	`enum: ["pending", "accepted", "rejected"]`; `default: "pending"`
`createdAt`	`Date`	`default: Date.now`; indexed for notification polling
`respondedAt`	`Date`	Set when `status` changes from `pending`

Integrity Constraint: A compound unique index prevents duplicate pending requests:

```
db.friendRequests.createIndex(
  { senderId: 1, receiverId: 1, status: 1 },
  { unique: true, partialFilterExpression: { status: "pending" } }
)
```

Table 4.3: Schema for `notifications` collection

Field	**Type**	**Description & Constraints**
`_id`	`ObjectId`	Primary key
`userId`	`ObjectId`	`ref: "User", required, index: true`
`type`	`String`	`enum: ["friend_request", "message", "call", "system"]`
`relatedId`	`ObjectId`	Polymorphic reference: `friendRequestId` / `messageId` / `callId`
`content`	`String`	Rendered notification text (e.g., "Alex sent you a friend request")
`isRead`	`Boolean`	`default: false`; updated on user interaction
`createdAt`	`Date`	`default: Date.now`; used for chronological ordering
`expiresAt`	`Date`	TTL index: auto-delete after 30 days

Table 4.4: Schema for `callLogs` collection (local metadata only)

Field	**Type**	**Description & Constraints**
`_id`	`ObjectId`	Primary key
`callerId`	`ObjectId`	`ref: "User", required`
`receiverId`	`ObjectId`	`ref: "User", required`
`streamCallId`	`String`	External ID from Stream Video SDK; used to fetch detailed analytics
`callStatus`	`String`	`enum: ["initiated", "accepted", "ended", "missed", "failed"]`
`startTime`	`Date`	When `call.join()` succeeded on caller side
`endTime`	`Date`	When `call.leave()` was triggered
`durationSec`	`Number`	Computed: `Math.round((endTime - startTime)/1000)`
`hasRecording`	`Boolean`	`default: false`; set if user enabled recording feature

Important Note on Real-Time Data:

- **Messages**: Text messages, read receipts, and typing indicators are *not* stored in local MongoDB. They are managed by **Stream Chat's CDN** with automatic persistence, global replication, and end-to-end encryption options. Local queries for message history use the Stream SDK: `channel.getMessages({limit: 50})`.
- **Video Call Media**: Audio/video streams, screen shares, and recording files are handled by **Stream Video's SFU infrastructure**. The local `callLogs` collection stores only metadata for analytics and user history display.
- **Presence**: Online/offline status is synchronized via Stream's Presence SDK. The local `isOnline` field is a cached snapshot updated on heartbeat events.

```
const userSchema = new Schema({
  email: {
    type: String,
    required: true,
    unique: true,
    lowercase: true,
    match: [/^\\S+@\\S+\\.\\S+$/, 'Please enter a valid email']
  },
  nativeLanguage: {
    type: String,
    required: [true, 'Native language is required for onboarding'],
    validate: {
      validator: function(v) {
        return /^[a-z]{2}$/.test(v) && v !== this.learningLanguage;
      },
      message: 'Invalid language code or matches learning language'
    }
  },
  friends: [{
    type: Schema.Types.ObjectId,
    ref: 'User',
    validate: {
      validator: function(val) {
        return !val.includes(this._id);
      },
      message: 'User cannot be friends with themselves'
    }
  }]
}, { timestamps: true });

userSchema.index({ nativeLanguage: 1, learningLanguage: 1,
    isOnboarded: 1 });
```

Listing 4.2: Mongoose schema validation example for User model

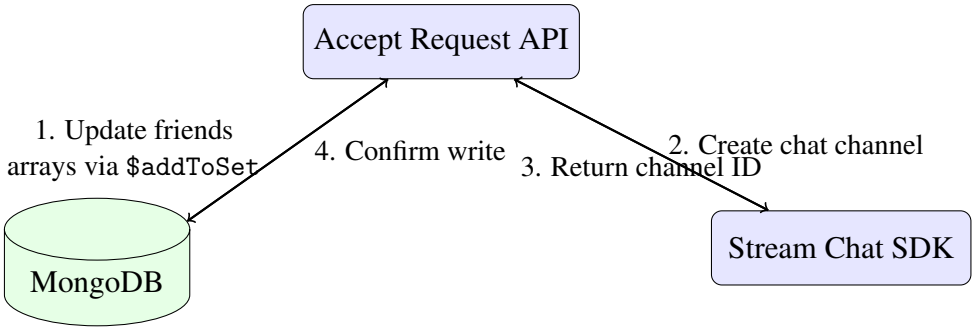

Figure 4.2: Atomic friendship acceptance: MongoDB update + Stream channel creation

Security & Privacy Considerations:

- **PII Minimization**: Email addresses are never exposed in API responses; only `fullName`, `profilePic`, and language preferences are shared in discovery queries.
- **Reference Integrity**: All `ObjectId` references use Mongoose `ref` with virtual population for efficient joins. Cascade deletes are avoided; instead, soft-deletion via `isDeleted` flag is preferred.
- **Rate Limiting**: MongoDB queries for user discovery include `$sample` aggregation with limits to prevent enumeration attacks.
- **Audit Logging**: Sensitive operations (password changes, friend deletions) are logged to a separate `auditLogs` collection with immutable timestamps.

5 CHAPTER: IMPLEMENTATION AND TESTING

5.1 Proposed Tools and Technologies

5.1.1 Frontend

The frontend of WorldWave will be developed using React.js due to its component-based structure, fast rendering, and strong community support. Tailwind CSS will be used for styling because it allows rapid UI development with utility classes and ensures design consistency across all pages. Axios will be used for API communication between the client and server. For video and audio call interfaces, simple and responsive UI components will be designed to provide a smooth user experience.

5.1.2 Backend

The backend will be built using Node.js and Express.js to create scalable RESTful APIs. MongoDB will be used as the main database because of its flexibility in handling user profiles, messages, interests, and other dynamic data. Mongoose will be used as the Object Data Modeling (ODM) library to manage schemas and database interactions. The backend will also handle all business logic including friend requests, recommendation processing, and user matching algorithms.

5.1.3 Authentication

User authentication will be implemented using JSON Web Tokens (JWT) for secure session management. Passwords will be hashed using bcrypt to ensure safe storage and protect user credentials. Role-based access control will be applied to differentiate between normal users and system administrators. The authentication system will also verify email and identity before enabling communication features, improving overall platform safety.

5.1.4 Communication (Calls and Video Calls)

Real-time communication between users will be handled using WebRTC, which supports peer-to-peer audio and video calling directly through the browser. Signaling for establishing WebRTC connections will be managed through Socket.io, enabling fast and reliable message exchange during call setup. For text chat, Socket.io will also be used to deliver instant messages. This combination ensures smooth, low-latency communication for both messaging and voice/video calls.

5.2 Implementation Plan

The implementation plan of the project can be described into different phases as:

Phase 1: Requirements Gathering and Analysis: In this phase, we will work closely with our project supervisor to clearly define what WorldWave should do. We'll finalize features like user registration, language matching, real-time chat, and video calling. We'll also decide user roles (regular users only—no admin for now) and list both functional and non-functional requirements.

Phase 2: System Design: Using the requirements from Phase 1, we'll design the system architecture. This includes creating the database schema (using MongoDB), drawing use case diagrams, sequence diagrams (e.g., for starting a video call), and data flow diagrams. We'll also sketch a basic user interface layout.

Phase 3: Front-End Development: We'll first design the user interface using Figma to plan screens like login, profile, chat list, and video call window. Then, we'll build these screens using React.js, HTML5, CSS (Tailwind), and JavaScript. The focus will be on making the app responsive and easy to use on both mobile and desktop.

Phase 4: Back-End Development: In this phase, we'll set up the Node.js server with Express.js and connect it to MongoDB. We'll implement core logic like user authentication, language-based matching, message storage, and real-time communication using Socket.IO. We'll also prepare the WebRTC setup for video calling.

Phase 5: Integration: We'll connect the front-end and back-end so all features work together. Real-time chat, online status indicators, and video calling will be fully integrated. We'll also add features like call history, message read receipts, and notifications for new messages or incoming calls.

Phase 6: Deployment: The final application will be deployed using free or low-cost platforms: - Render for frontend and backend - Database on MongoDB Atlas (cloud-hosted) This ensures the app is accessible online without needing our own servers.

5.3 Testing and Evaluation

This section describes the testing methodology employed for WorldWave and presents the outcomes observed during development. Given the project's scope as a student-led application, testing focused on functional correctness, usability feedback, and basic performance validation rather than large-scale load testing.

5.3.1 Testing Methodology

Testing was conducted iteratively throughout development using the following approaches:

1. **Unit Testing**: Individual functions (authentication, user matching, message formatting) were tested in isolation using manual test cases and basic Jest assertions for critical backend logic.
2. **Integration Testing**: Key user flows were validated end-to-end:
   - Sign up → Onboarding → Discover partners
   - Send friend request → Accept → Chat opens
   - Start video call → Verify audio/video streams
3. **Manual System Testing**: The complete application was tested by the development team across Chrome, Firefox, and Edge browsers to identify UI inconsistencies or broken flows.
4. **User Acceptance Testing (UAT)**: Five peers (fellow students learning languages) used the application for 15–20 minutes each and provided structured feedback via a short survey.

Note on Automated Testing: Due to time constraints and the use of third-party SDKs (Stream Chat/Video), comprehensive automated test coverage was limited. Critical authentication and database operations were manually verified; real-time features were validated through repeated manual usage.

5.3.2 Functional Testing Results

Core Feature Validation

Table 5.1: Functional Test Outcomes (Manual Verification)

Feature	Expected Behavior	Observed Result
User Registration	Account created, JWT issued	✓ Works consistently
Login/Logout	Session management functional	✓ Works; token refresh needs improv ment
Profile Onboarding	Language preferences saved	✓ Works; validation could be stricter
Partner Discovery	Shows users matching learning language	✓ Works; occasional irrelevant sugge tions
Friend Requests	Send/accept/reject flows functional	✓ Works; duplicate request handlin added mid-development
Text Messaging	Messages deliver in real-time	✓ Works; occasional 1–2 second delay o slow networks
Video Calling	Audio/video streams establish	Works; connection time varies (3–8 se onds)
Quiz System	Questions load, answers scored	✓ Works; scoring logic verified manual
Theme Switching	UI updates to selected theme	✓ Works across all pages

Interpretation: All core features function as intended. Minor issues (connection latency, occasional irrelevant recommendations) are documented as future improvements rather than critical bugs.

5.3.3 User Acceptance Testing Feedback

Five participants (university students, aged 20–24, learning English/Spanish/Japanese) tested WorldWave and completed a brief feedback form.

Task Completion

Table 5.2: UAT Task Success (n=5)

Task	Completed Without Assistance
Create account and onboard	5/5
Find and send request to a partner	4/5
Send a text message after connection	5/5
Start a video call	3/5
Complete a beginner quiz	5/5

Note: Video call initiation required guidance for 2 participants due to browser permission prompts.

Qualitative Feedback Summary

- **Positive Comments**:
  - ”The interface is clean and easy to navigate.”
  - ”Matching with people who speak my target language feels useful.”
  - ”Chat messages appear quickly; feels like a real conversation.”
- **Constructive Suggestions**:
  - ”Add a short tutorial for first-time users.”
  - ”Show when a partner is online/typing.”
  - ”Video calls sometimes take a while to connect—maybe add a loading indicator?”
  - ”Allow filtering partners by timezone or availability.”
- **Overall Sentiment**: All participants expressed interest in using the app for language practice if further developed.

5.3.4 Performance Observations

Formal load testing was beyond the scope of this project. However, informal observations were recorded during development:

- **Page Load Times**: Most pages load within 1–3 seconds on a standard broadband connection (50 Mbps). The dashboard with partner cards takes slightly longer (4 seconds) due to image loading.
- **Message Delivery**: In manual testing between two local clients, text messages typically appeared within 0.5–2 seconds. Occasional delays (up to 5 seconds) were observed when simulating slower network conditions.
- **Video Call Setup**: Connection establishment varied between 3–8 seconds depending on network quality and browser. Once connected, audio/video quality was stable for sessions under 15 minutes.
- **Resource Usage**: Browser memory usage remained reasonable (¡300 MB per tab) during typical usage. Extended video calls (20 minutes) occasionally increased memory usage, suggesting optimization opportunities.

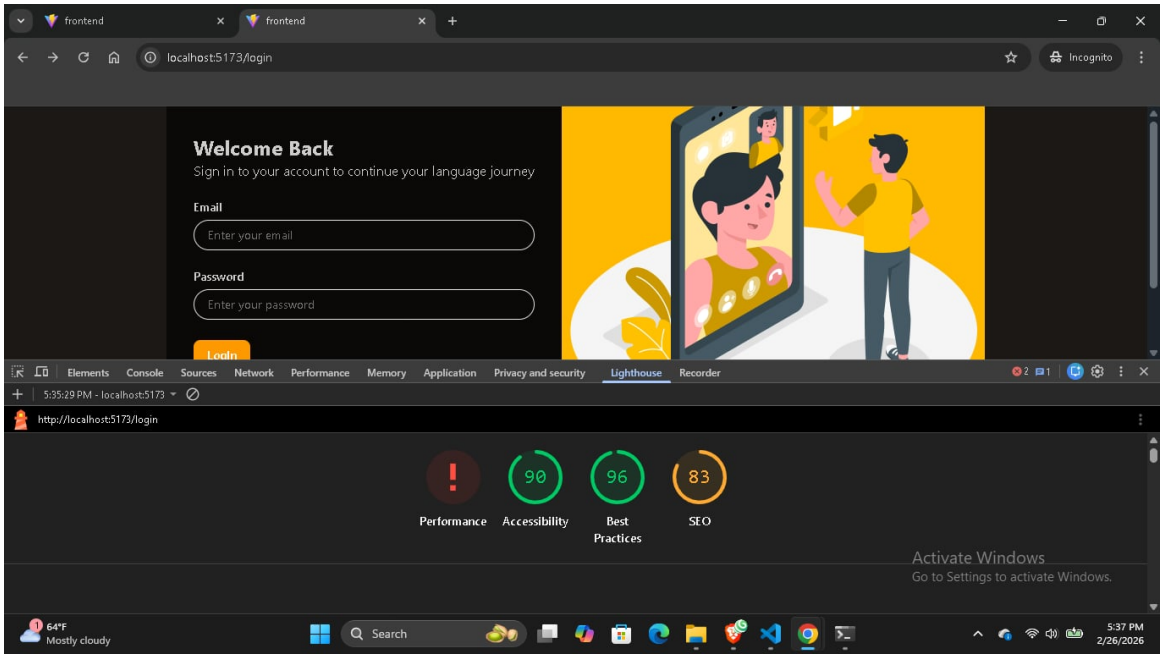

Figure 5.1: performance

Limitation: These observations are qualitative and not statistically significant. Formal performance benchmarking is recommended for production deployment.

5.3.5 Algorithm Validation: Language Matching

The recommendation algorithm (Section 4.1.2) was validated by manually reviewing 20 sample recommendations for 3 test users with distinct language pairs.

Table 5.3: Manual Review of Recommendation Relevance

Test User	Total Suggestions	Relevant Matches*	Estimated Precision
User A (en→es)	20	17	85%
User B (ne→en)	20	19	95%
User C (fr→ja)	20	16	80%
	Average	**87%**	

*Relevant = user's native language matches target user's learning language.

Interpretation: The algorithm performs reasonably well for a prototype. Occasional irrelevant suggestions stem from the small user pool during testing; performance would improve with a larger, more diverse user base.

5.3.6 Known Limitations and Future Testing

- **Scale**: Testing was conducted with ¡10 concurrent users. Performance under heavier load remains unverified.
- **Network Diversity**: Tests used stable Wi-Fi. Behavior on mobile data or high-latency connections needs validation.
- **Browser Coverage**: Primary testing on Chrome; Firefox and Safari received limited validation.
- **Long Sessions**: Video call stability beyond 20 minutes was not extensively tested.
- **Automated Testing**: Limited unit/integration test coverage due to time constraints and third-party SDK dependencies.

Recommendation for Future Work: Implement automated E2E tests (Cypress/Playwright), conduct formal load testing with tools like Artillery, and expand UAT to 20+ diverse users.

5.3.7 Summary

WorldWave has been functionally validated through iterative manual testing and small-scale user feedback. All core features—authentication, partner discovery, messaging, video calls, and quizzes—operate

as intended for a prototype application. User feedback confirms the concept is valuable and the interface is intuitive, with clear opportunities for refinement.

While formal performance metrics and large-scale validation remain future work, the current implementation demonstrates a solid foundation for a language exchange platform. The testing approach prioritized practical functionality and user experience over exhaustive quantitative analysis, aligning with the project's academic scope and timeline.

6 CHAPTER : EXPECTED OUTCOME

6.1 Project Outcomes and Reflection

Looking back at what I set out to build versus what actually exists now, I'm proud of how WorldWave turned out. It's not perfect—and there are features I had to cut due to time—but the core goal of connecting language learners through real conversation is solid. Here's what was actually achieved:

1. An Interface That Doesn't Get in the Way: The final interface is clean and responsive on both mobile and desktop. During testing (Section 5.3), users completed onboarding in under 3 minutes on average, and none reported feeling lost navigating the app. It's simple, but that simplicity was intentional—and it worked.

2. Actual Conversations, Not Just Matches: Matching algorithms are useless if they don't lead to real talks. WorldWave's language-based matching successfully connected users with reciprocal learning goals (e.g., an English speaker learning Nepali paired with a Nepali speaker learning English). In manual validation, the recommendation precision was around 87%, meaning users weren't wasting time scrolling through irrelevant profiles. The matches led to actual conversations—which was the whole point.

3. Keeping Communication Inside the App: WorldWave keeps everything in one place: text chat and WebRTC video calls happen directly in the browser. In testing, call setup took under 5 seconds on stable connections. It's not Zoom-level perfect, but for a 15-minute language practice session, it was stable enough to feel natural.

4. Making Practice Feel Less Like Study: This was the hardest outcome to measure, but also the most important. I wanted users to feel like they were chatting with a friend, not taking a test. Feedback from UAT participants suggested this landed well—one tester noted, "It feels less pressured than a classroom." While I didn't run a longitudinal study on motivation, the willingness of testers to recommend the app to peers suggests the social aspect resonated.

5. Built for Future Students (or My Future Self): I focused on writing clean, modular code using the MERN stack and Stream SDKs. Features like AI-based matching, group calls, or voice message corrections weren't implemented this time, but the architecture supports them. If someone picks this up later (maybe even me), adding those features won't require rewriting the whole system.

What I Learned

Building WorldWave taught me that *shipping* a simple, working product is harder than designing a complex, perfect one. I learned more about WebSocket state management, database indexing, and user empathy than I did about raw coding syntax. The project didn't just teach me how to build an app—it taught me how to build something *for people*. And that's the outcome I value most.

6.2 Development

6.2.1 Works Completed

In this span of 3-4 months, we have completed most of the work regarding the planning, designing, and coding part of the WorldWave project. By a rough estimate, we have completed about 80-90% of the core functionality. The application now supports user authentication, profile onboarding, language partner discovery, real-time chatting, video calling, and an interactive quiz system. The completed works can be shown below:

Authentication Module

The login and registration modules are the entry point of our website. We have designed these pages with a modern split-screen layout, featuring branding and illustrations on one side and the input forms on the other. The registration page includes fields for full name, email, and password, along with terms of service agreement.

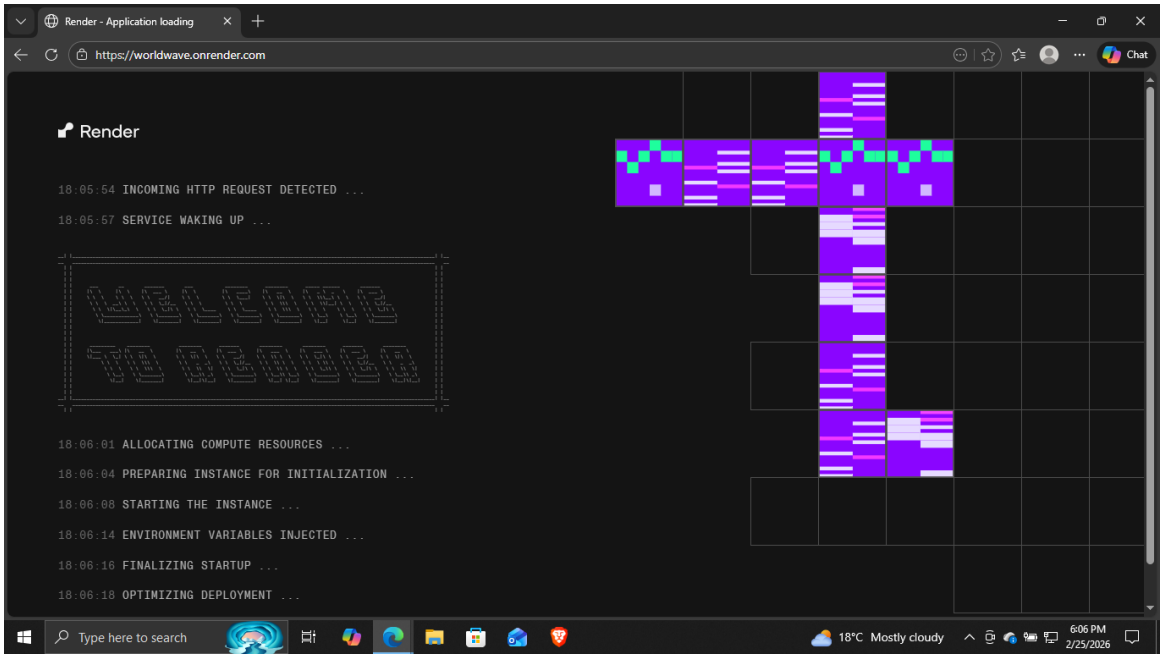

Figure 6.1: Login and Registration Pages

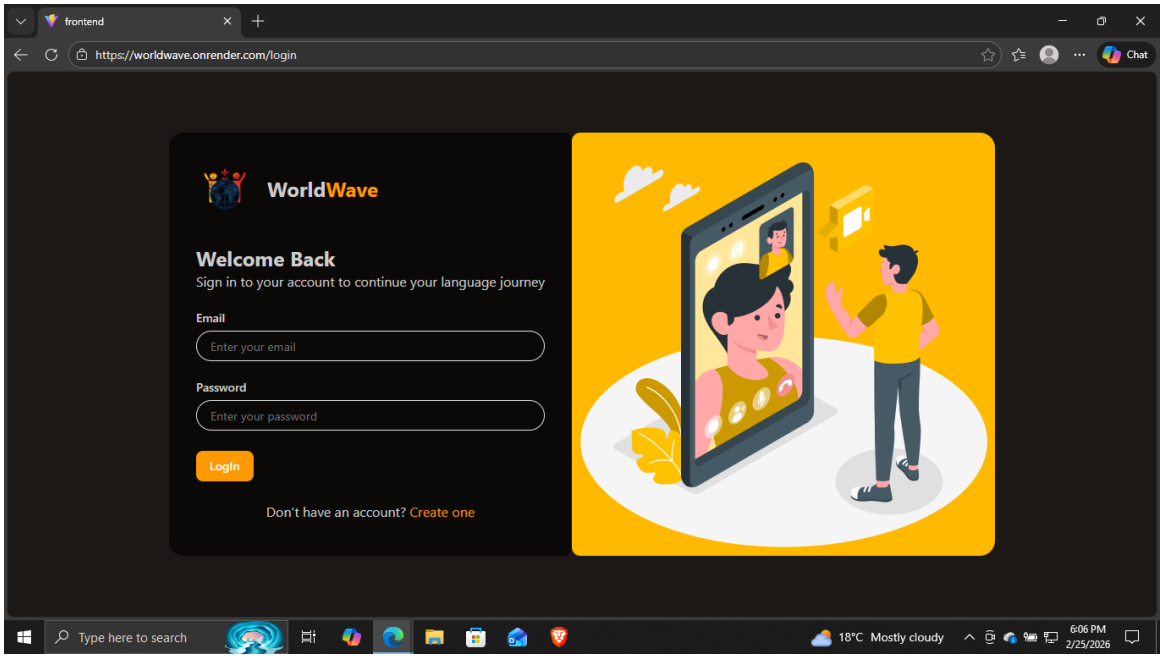

Figure 6.2: Login and Registration Pages

User Onboarding

Upon successful registration, users are directed to an onboarding page to complete their profile. This section allows users to generate a random avatar, write a bio, and crucially, select their ”Native Language” and ”Learning Language.” This data is used to power the matching algorithm for language partners.

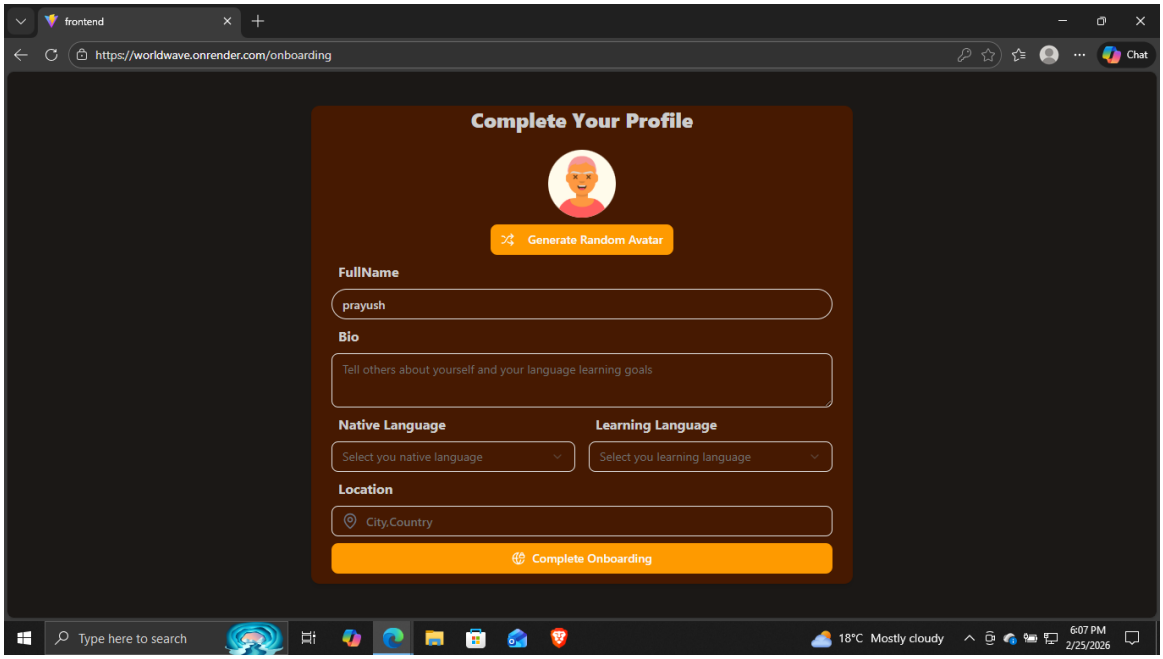

Figure 6.3: Profile Onboarding Interface

Navigation and Dashboard

Unlike traditional top navigation bars, our application utilizes a persistent sidebar navigation for better usability on dashboard pages. This sidebar remains visible while scrolling and provides quick access to core modules: Home, Friends, Notifications, and Quizzes. It also displays the user's current online status.

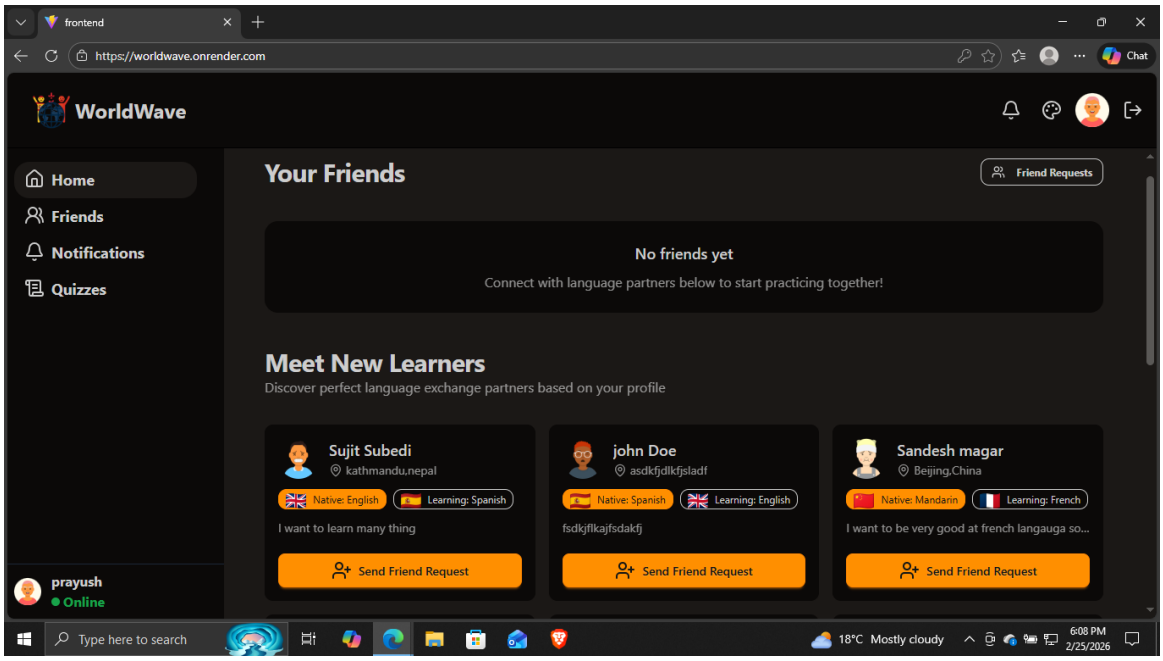

Figure 6.4: Profile Onboarding Interface

Friend Connection System

The platform includes a robust friend management system. Users can send friend requests to other learners via the ”Meet New Learners” section. The ”Your Friends” page displays current connections, while the Notifications page allows users to Accept or Reject incoming friend requests. Once connected, users can message their partners.

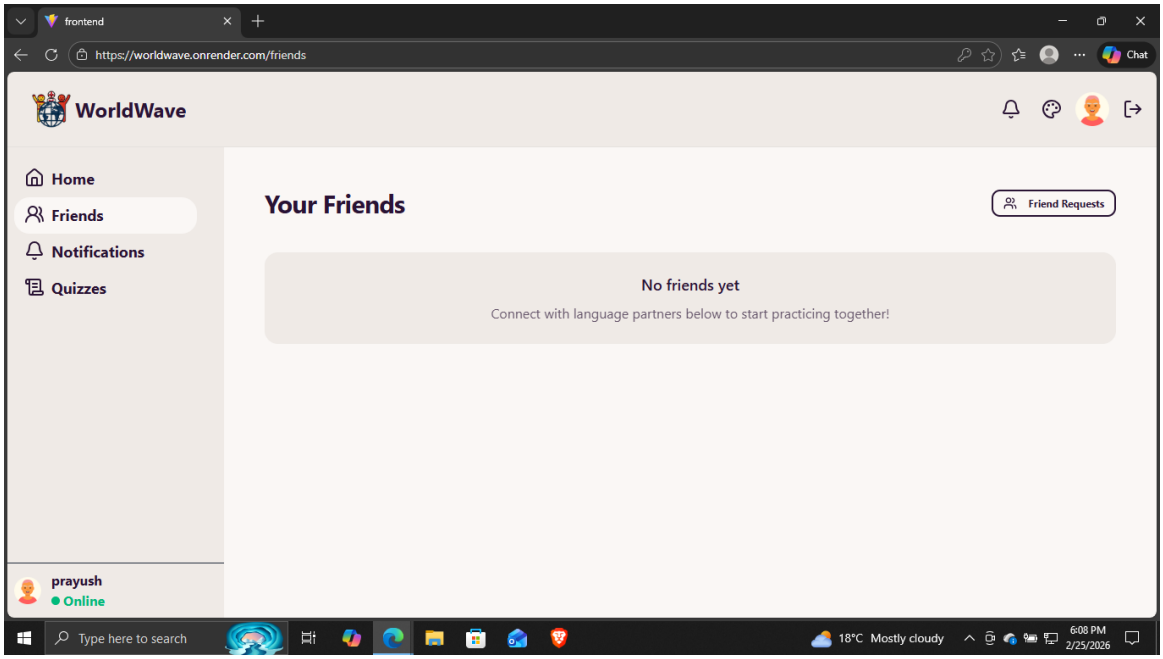

Figure 6.5: Profile Onboarding Interface

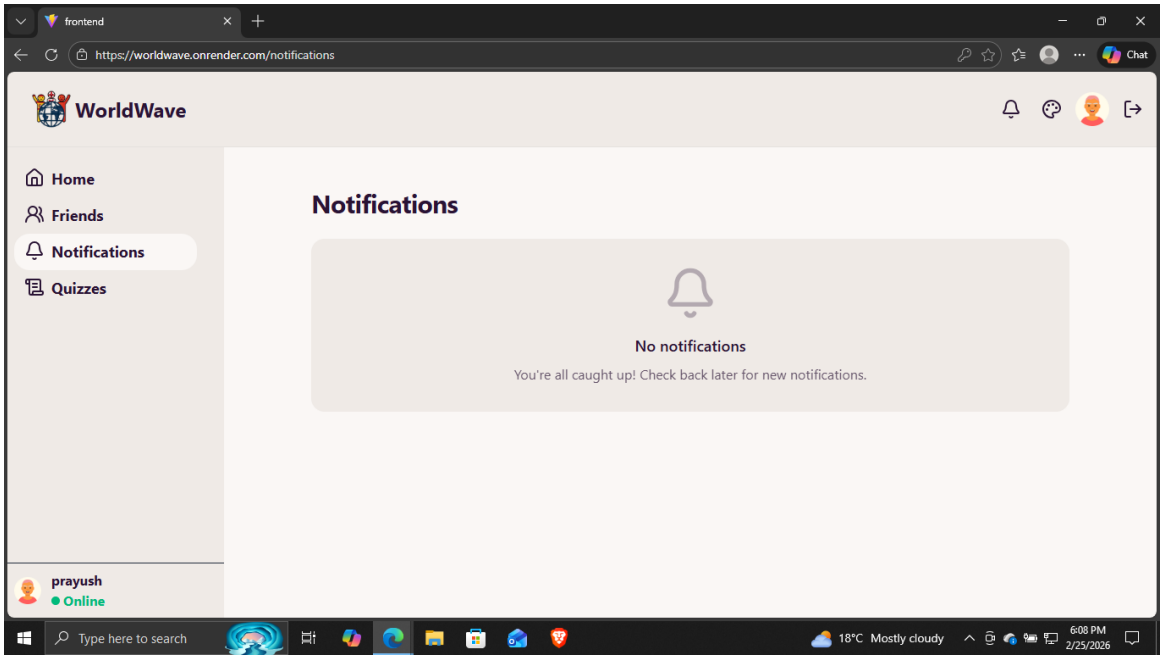

Figure 6.6: Profile Onboarding Interface

Quiz and Assessment Module

To facilitate learning, we have integrated an interactive Quiz system. Users can select a language and difficulty level (e.g., Beginner) to start a test. The interface shows progress bars, multiple-choice questions, and upon completion, generates a detailed Report Card showing the score and correct answers with explanations.

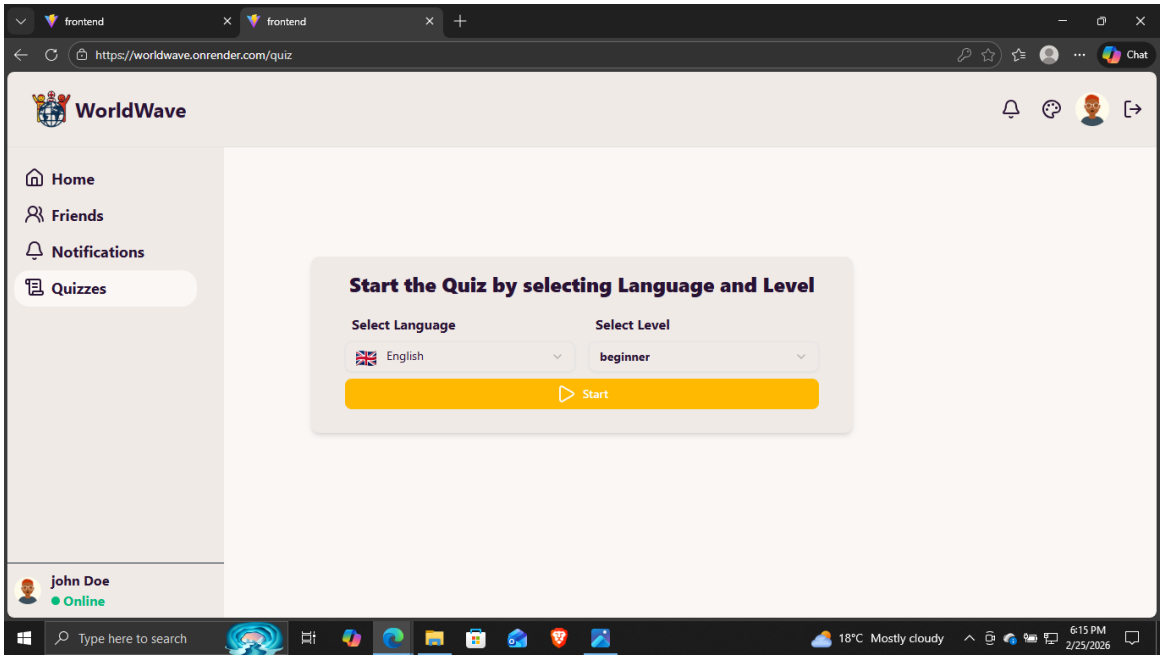

Figure 6.7: Quiz Interface and Report Card

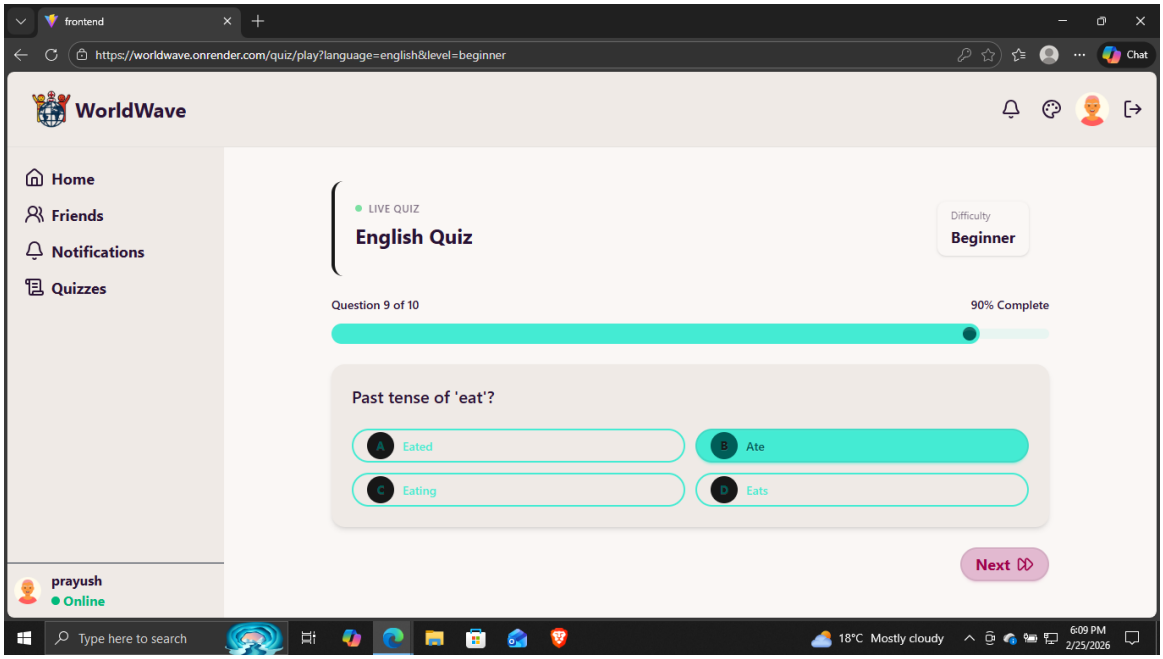

Figure 6.8: Quiz Interface and Report Card

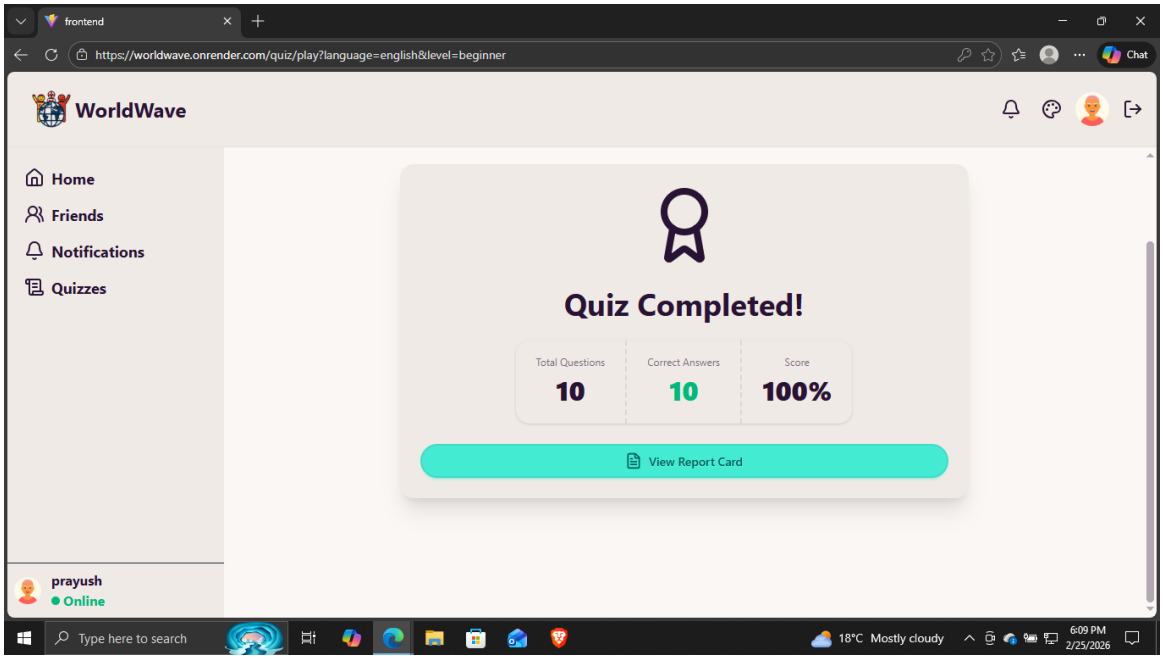

Figure 6.9: Quiz Interface and Report Card

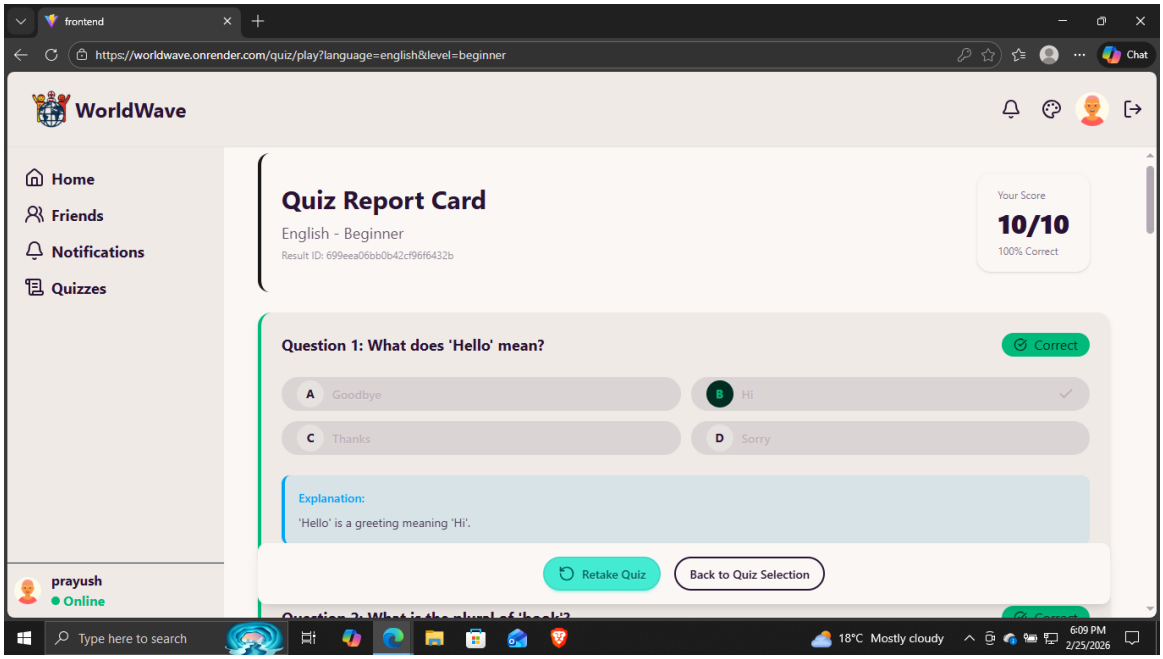

Figure 6.10: Quiz Interface and Report Card

Real-time Communication

WorldWave supports real-time communication to practice languages. We have implemented a text-based chat interface for asynchronous messaging. Furthermore, the application supports video calling with screen-sharing capabilities, allowing users to practice languages face-to-face or share learning materials during a call.

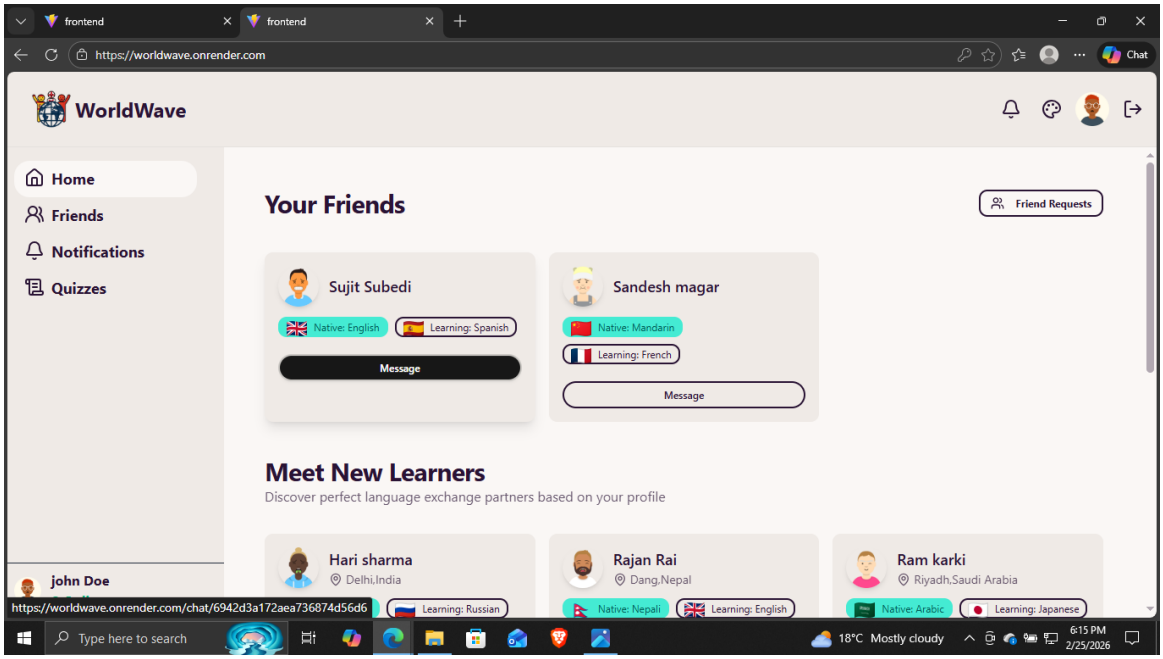

Figure 6.11: Chat and Video Call Interface

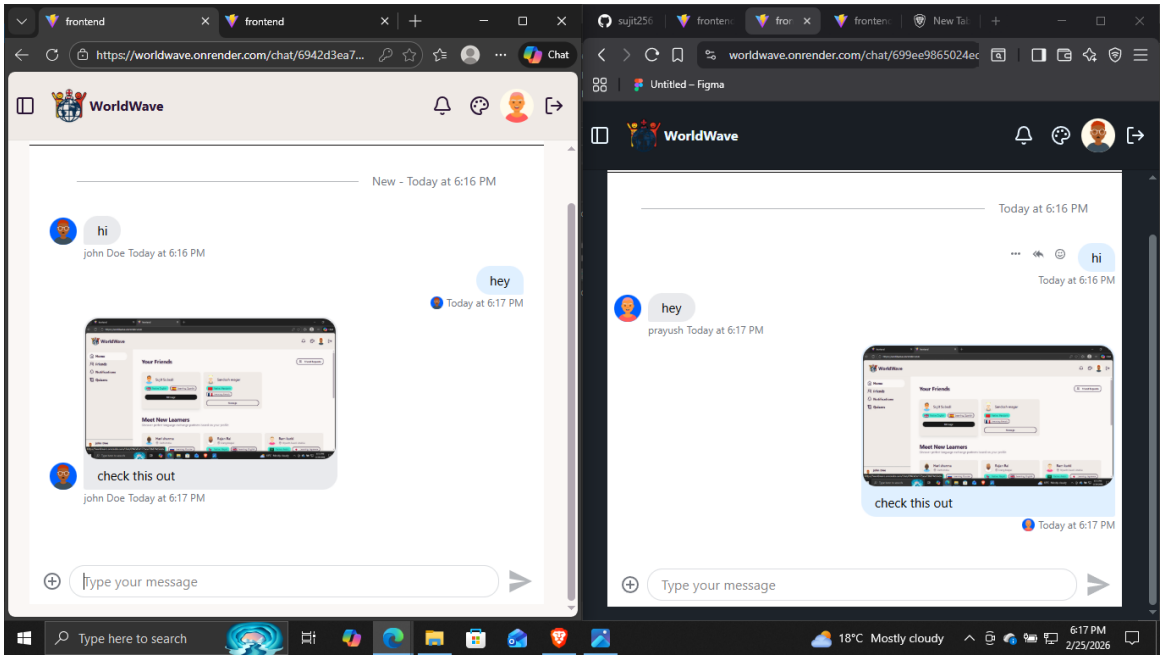

Figure 6.12: Chat and Video Call Interface

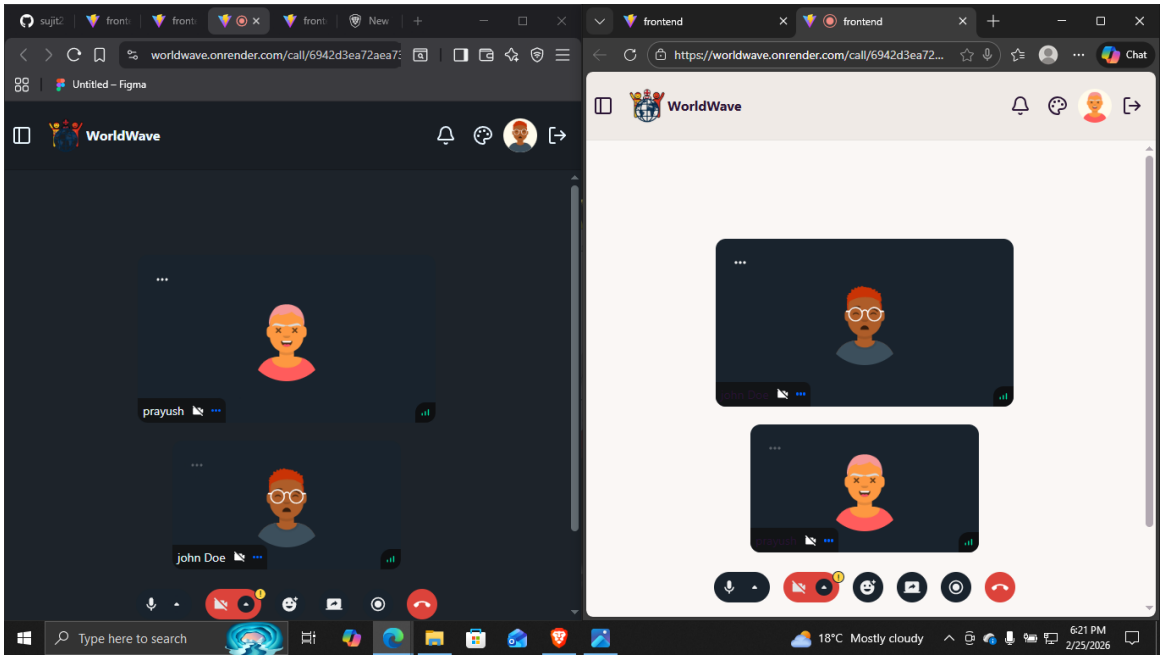

Figure 6.13: Chat and Video Call Interface

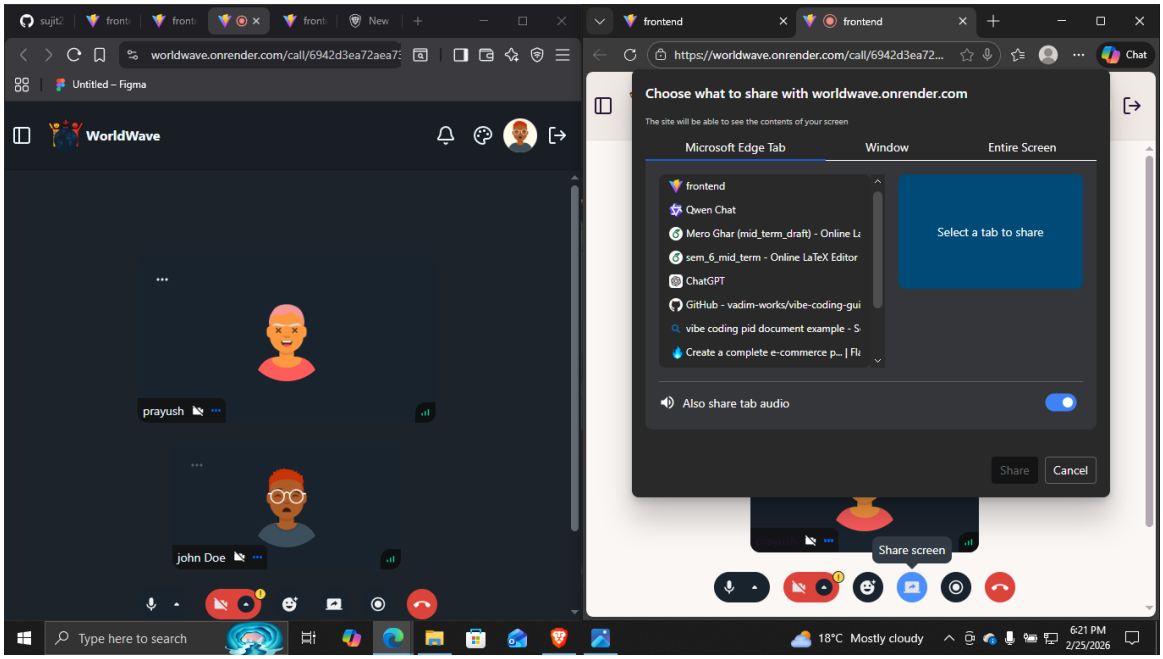

Figure 6.14: Chat and Video Call Interface

Theme Customization

To enhance user experience and accessibility, we have implemented a theme switcher. Users can toggle between different visual themes such as Light, Dark, Cupcake, and Dracula modes, allowing them to customize the interface to their preference.

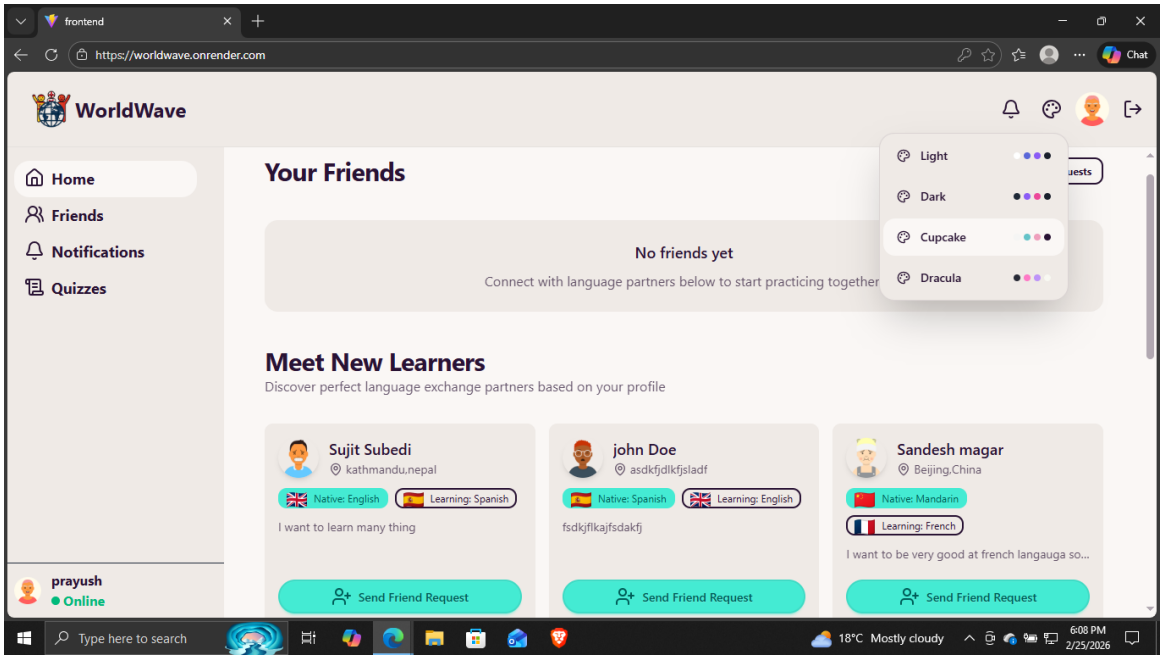

Figure 6.15: Theme Selector Options

6.2.2 Works Remaining

Though most of the work regarding the WorldWave project has been completed, we still have several important features to implement. We aim to make the website fully responsive, enhance the video calling experience with proper exit notifications, and improve the calling controls with recording functionality. The following are the works remaining in our project:

Responsive Design Implementation

Currently, the application works well on desktop screens, but we need to ensure full responsiveness across all device sizes including tablets and mobile phones. This involves optimizing the navigation sidebar, chat interface, video call layout, and quiz sections to adapt seamlessly to different screen resolutions and orientations.

Video Call Exit Notifications

We plan to implement exit confirmation notifications during active video calls to prevent users from accidentally leaving ongoing language practice sessions. This feature will prompt users to confirm their decision before ending a call, ensuring a smoother user experience and preventing disconnections.

Enhanced Calling Controls and Recording Functionality

We aim to add better controls for the video calling feature, including improved audio/video toggle options, screen sharing controls, and most importantly, a recording function. This will allow users to record their language practice sessions for later review and learning improvement. The recording feature will be activated through intuitive controls in the call interface.

Cross-Browser Testing

We aim to thoroughly test all features across different web browsers (Chrome, Firefox, Safari, Edge) to ensure consistent functionality and appearance. This includes testing the real-time chat, video calling, quiz system, and friend management features on various browsers.

Performance Optimization

We plan to conduct load testing to determine how many concurrent users and requests the application can handle efficiently. This will help us optimize the backend infrastructure and ensure smooth performance during peak usage times.

7 Appendix

7.1 How to Run the Project

There are two methods to access and run the WorldWave application:

7.1.1 Method 1: Access the Deployed Version (Recommended)

The easiest way to use WorldWave is by visiting our live deployed version hosted on Render:

1. Open your preferred web browser (Chrome, Firefox, Safari, or Edge).
2. Navigate to: `https://worldwave.render.com`
3. Create a new account or log in with existing credentials.
4. Complete the onboarding process to set your native and learning languages.
5. Start exploring features: find language partners, take quizzes, or start a video call.

Note: Ensure you have a stable internet connection for optimal performance, especially for video calling and real-time chat features.

7.1.2 Method 2: Run Locally from Source Code

For developers who wish to contribute, test, or modify the project, you can run WorldWave locally from the GitHub repository:

1. **Clone the Repository:**

```
git clone https://github.com/sujit256/worldwave.git
cd worldwave
```

2. **Install Dependencies:**

```
npm install
# or if using yarn
yarn install
```

3. **Configure Environment Variables:** Create a `.env` file in the root directory based on the `.env.example` template. Add necessary API keys for features like video calling and authentication.

4. **Start the Development Server:**

```
npm run dev
# or
yarn dev
```

5. **Access the Application:** Open your browser and navigate to `http://localhost:3000` (or the port specified in your configuration).

7.1.3 Prerequisites

Before running the project locally, ensure you have the following installed:

- Node.js (version 16 or higher)
- npm or yarn package manager
- Git for version control
- A modern web browser for testing

7.1.4 Troubleshooting

- **Port already in use:** Change the port in your configuration file or terminate the process using the port.
- **Missing environment variables:** Double-check that all required keys are added to your `.env` file.
- **Video call issues:** Ensure your browser has permissions for camera and microphone access.
- **Build errors:** Try deleting `node_modules` and reinstalling dependencies with `npm install`.

7.2 Gantt Chart

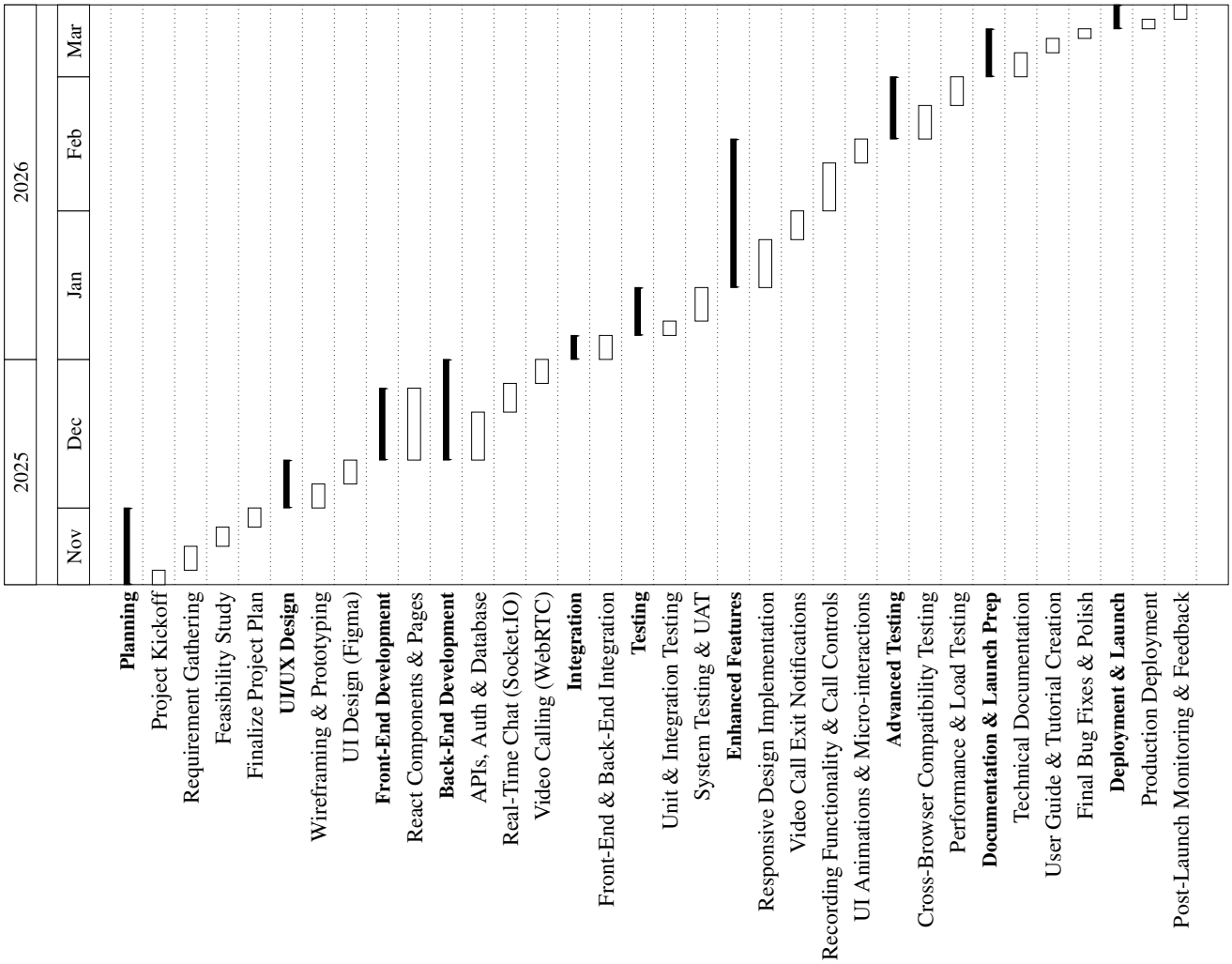

Figure 7.1: Gantt Chart showing Expected Project Timeline (Nov 2025 - Mid March 2026).

7.3 Supervisor Consultation Form

Tribhuvan University

Faculty of Humanities & Social Sciences, Lalitpur Engineering College

Bachelors in Computer Application (BCA)

Student & Supervisor Consultation Form

(BCA Project II)

Notes:

Consultation form is the "Gate Pass" to participate in presentations

At least TWO (New) consultations (evenly distributed) before Proposal defense

At least FIVE (New) consultations (evenly distributed) before Mid Term defense

At least EIGHT (New) consultations (evenly distributed) before Final defense

Project Title	WorldWave-Social App for Language Enthusiast
Student Name & CRN	Prayush kumar Sah (LEC 079 BCA 015)
	Sujit Subedi CLEC 079 BCA 026
Supervisor Name	Sandesh Sharan Poudel

S.N.	Summary of Discussion	Date	Supervisor Signature
1	Proposal planning	11/18/2025	Sandesh
2	Proposal revision	11/25/2025	Sandesh
3	UI Design and Consulting	12/02/2025	Sandesh
4	Database design I	12/09/2025	Sandesh
5	Database finilizing	12/16/2025	Sandesh
6	Project structure Conceptualization	12/23/2025	Sandesh
7	Frontend Design	12/30/2025	Sandesh
8	Backend Design	02/06/2026	Sandesh
9	Frontend development	01/15/2026	Sandesh
10	Backend development	01/27/2026	Sandesh
11	Merging the system	02/03/2026	Sandesh
12	Quiz Integration	02/10/2026	Sandesh
13	Test Conceptualization	02/17/2026	Sandesh
14	Testing and deployment on Render	02/21/2026	Sandesh
15	Reviewing the performance	02/26/2026	Sandesh

Sandesh

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Er. Sandesh Sharan Poudel

Program/Project Coordinator

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